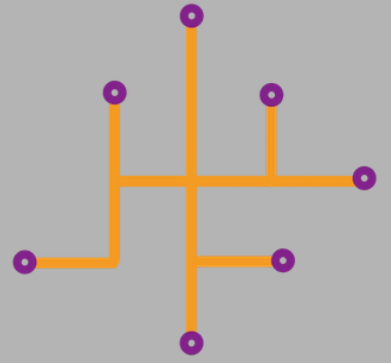




A Student's Perspective of the Australian Semiconductor Industry

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To the Semiconductor Sector Service Bureau (S3B) and the Australian Semiconductor Community,

As two undergraduate engineering students who are passionate about microelectronics and committed to building our careers within Australia, we welcome the release of the National Semiconductor Roadmap Discussion Paper. We believe that the future of the semiconductor industry in Australia is bright, that student interest in hardware and microelectronics is plentiful, and that industry demand for a future talent pipeline is growing. However, Australia currently lacks the framework required to nurture this interest at universities and to have industry effectively absorb new graduates into the domestic ecosystem.

We offer this letter as a thought piece during the S3B's engagement phase to share our undergraduate experience and help shape Australia's first National Semiconductor Roadmap.

The Current University Landscape and Student Perception

An Australian undergraduate does not have a coherent pathway for semiconductor engineering. Instead, relevant knowledge is fragmented across broader Electrical and Computer Engineering degrees, delivered primarily through isolated electives and research projects.

University courses remain heavily theoretical and lack practical grounding. While we are taught circuit design and device physics, we are seldom taught the details of how those components are physically fabricated, packaged, and tested in industry. Core topics like digital design are often delayed until third year, long after introductory courses in embedded systems, general software, and traditional electronics have drawn talent away. Without guidance to



inform self-driven projects, such as a clear progression from FPGA platforms to verification and ASIC design, many interested students stagnate and lose momentum.

Among the broader student body, there is a prevailing belief that local semiconductor opportunities are virtually non-existent. This misconception is often reinforced by academic staff, despite the reality of Australia's growing fabless design, photonics, and quantum ecosystems. The common consensus becomes that to have a successful career in microelectronics, one must go overseas.

Our own experiences, such as participating in S3B semiconductor bootcamps and visiting overseas hubs like Taiwan, demonstrate that this narrative is incomplete. We believe Australia cannot and should not attempt to compete on high-volume silicon manufacturing or advanced packaging at scale. Instead, our strength lies in specialised design-led niches, advanced materials, and application-specific integration. Capitalising on these niches requires cultivating domestic talent while welcoming experienced international professionals.

As students, we feel frustrated by the lack of direct federal support for early-career practical experience. Securing funding for hands-on extracurriculars, international study programs, or physical tapeout opportunities remains extraordinarily difficult for an individual alone to attain. While S3B has done vital work bridging industry gaps at the state level in NSW, a national platform backed by federal support is essential to shift student perceptions nationwide. We can clearly see the tangible effects of this support already; students in New South Wales are noticeably more invested and aware of the industry compared to their peers in the ACT or Queensland, where this centralised presence is



currently absent. A nationalised S3B would scale this successful model, strengthening the university network and ensuring that geography does not dictate a student's exposure to our growing ecosystem.

Recommendations for the Student Pipeline

Right now, this industry only finds students who go looking for it. That isn't a pipeline, it's chance. We believe that Australia's semiconductor student future must be built by design, not by discovery.

To convert student interest into a sustainable domestic workforce, we should change how and when students engage with this field. For example, introducing digital design and semiconductor physics concepts earlier in undergraduate degrees, ideally within the first two years, to capture talent before students commit to other engineering streams. This early engagement must be paired with access to professional tooling. Currently, universities like USYD and UNSW introduce commercial Electronic Design Automation software, such as Cadence and Synopsys, in fourth-year courses. This is simply too late. While some earlier courses attempt to bridge this gap with open-source tooling, these alternatives lack the industry parity required to build competitive skillsets.

Early, deliberate, and continuous exposure is therefore critical. A path to this could be lobbying for group educational licenses that make professional tools viable for younger students. Crucially, if students lose access to these tools once a semester ends, the knowledge fades. Structures in university degrees and S3B initiatives should therefore aim to empower university student groups to undertake self-driven, extracurricular projects that utilise these tools year-round, cementing their early-career skillsets.



The semiconductor pipeline also requires diverse capabilities. We see TAFE and technical universities as an important training ground specifically for teaching semiconductor technician roles, adopting successful international models to support the realities of the domestic industry. Finally, to ensure these graduates are effectively absorbed into the ecosystem, we believe the government should develop absorption mechanisms. Federal subsidies that lower the financial barrier for local startups to take on new graduates and interns would help immensely in ensuring that there is a vibrant, welcoming local market ready for new talent.

Ultimately, these measures aim to cultivate a mainstream, default career path for Australian STEM talent, comparable to the mining or consulting sectors. By actively designing the semiconductor industry as a primary destination for engineering graduates, Australia can slow the brain drain and reposition itself to grow faster than before.

Australia offers an unmatched quality of life, and Australian students genuinely want to stay and contribute to building a sovereign, high-tech industry. New engineering graduates sit at a critical inflection point: deciding whether to stay and grow a nascent, high-potential ecosystem, or head offshore for postgraduate studies and employment.

We already possess innovative startups, world-class research institutions, and strong global connections. It is time to get serious about our national strategy and provide Australia's talented stream of students with the structure, tools, and opportunities needed to build our semiconductor future from home.